ITA0610-Machine Learning

LAB EXPERIMENTS

Vasantha Nithi D Y

192424137

Lab 33: Pandas File I/O (Reading and Writing CSV)

Aim

To read data from a **CSV file into a Pandas DataFrame**, process the data, and write the processed DataFrame back to a new CSV file.

Algorithm

1. Import the Pandas library.
2. Create sample student data.
3. Save the data as students.csv.
4. Read the CSV file using pd.read_csv().
5. Add a new column to identify whether the student has passed.
6. Save the processed DataFrame as students_processed.csv.
7. Display the processed data.

Program

```
# Lab 33: Pandas File I/O (Reading and Writing CSV)


import pandas as pd


# Create sample student data
data = {
   "name": ["A", "B", "C", "D"],
   "marks": [78, 45, 90, 62],
   "class": ["I", "II", "I", "II"]
}


df = pd.DataFrame(data)


# Save as CSV file
```

```
df.to_csv("students.csv", index=False)


# Read CSV file
df = pd.read_csv("students.csv")


print("Loaded data:\n", df)


# Process the data
df["pass"] = df["marks"] >= 50


# Save processed data
df.to_csv("students_processed.csv", index=False)


print("\nProcessed file saved. Preview:\n", df)
```

Sample Output

```
Loaded data:
  name  marks class
0    A     78     I
1    B     45    II
2    C     90     I
3    D     62    II

Processed file saved. Preview:
  name  marks class   pass
0    A     78     I   True
1    B     45    II  False
2    C     90     I   True
3    D     62    II   True
```

Result

Thus, **CSV file input and output operations were successfully performed using Pandas**, including reading, processing, and writing the data to a new CSV file.